Redis-Based Airbnb Data Management & Analytics for New York City

Any Das (922710) and Aarohi Mistry (925352)
CdLM Data Science, Università degli Studi di Milano-Bicocca

Abstract

This project presents the design and implementation of a high-performance Big Data pipeline for the New York City Airbnb market. By leveraging the in-memory capabilities of Redis, we transitioned from traditional static file analysis to a dynamic system capable of real time data retrieval and insight generation. The pipeline automatically ingests, cleans, and structures over thirty-six thousand listings into efficient Redis data models including Hashes, Sorted Sets, and Geospatial Indexes. Our subsequent business intelligence analysis revealed critical market trends such as a distinct dichotomy between luxury and budget friendly high rated listings and significant inventory shortages in residential neighborhoods like New Dorp. Furthermore, we identified a core set of amenities including smoke alarms and WiFi that are essential for guest satisfaction. This study demonstrates that modern NoSQL database solutions effectively transform raw hospitality data into actionable value for travelers, hosts, and platform administrators.

Keywords

Big Data Analytics, Redis, NoSQL Database, Airbnb, Data Pipeline, ETL, Business Intelligence, Python, Real-time Querying, New York City.

Table of Contents

1. Introduction **1**
2. Data Description **1**
3. Data Preprocessing & Cleaning **2**
4. Database Design with Redis **2**
5. Analysis and Key Insights **3**
6. Conclusion **5**
7. References **5**

1. Introduction

In the modern hospitality industry, data has become the most valuable asset. Platforms such as Airbnb generate massive volumes of unstructured data every day, including complex pricing changes and countless guest reviews. To make sense of this information, traditional static analysis is often insufficient. Consequently, this project focuses on designing and implementing a robust and automated Big Data Pipeline for the New York City Airbnb market by utilizing public datasets from Inside Airbnb [1].

The core objective of this work was to move beyond simple file processing and implement a high-performance database solution using Redis. By leveraging the in-memory capabilities of Redis, we aimed to build a system that stores data efficiently while allowing for real-time querying. This approach enables us to derive actionable business insights much faster than traditional methods.

Beyond the technical architecture, this solution is designed to deliver tangible value to key stakeholders in the rental ecosystem. For travelers, the system provides the capability to instantly rank properties based on value and specific amenities to ensure faster booking decisions. Simultaneously, it empowers hosts and property managers with deep insights into market dynamics. This includes identifying which neighborhoods command the highest prices and pinpointing areas with the highest occupancy demand. This dual focus ensures that the data pipeline serves both consumer needs and business strategy.

2. Data Description

To build a comprehensive picture of the New York City rental market, we utilized the public data repository provided by Inside Airbnb [1], capturing the market snapshot as of October 1, 2025. However, our approach went beyond simple data analysis; we engineered a fully automated pipeline capable of handling large-scale data ingestion. The pipeline automatically ingests three distinct datasets to ensure a multifaceted view of the market.

The Listings Data (*listings.csv.gz*) serves as the core component of our database. It contains detailed static attributes for 36,111 unique properties, featuring a rich schema with over 70 fields including the unique listing *id*, *neighbourhood_cleansed*, *room_type*, *price*, and *review_scores_rating*. We used this extensive feature set to populate the primary Redis Hashes and geospatial indexes effectively.

Complementing the static listings, the Calendar Data (*calendar.csv.gz*) represents the most voluminous part of our pipeline, containing over 13 million records (13,180,517 rows). It provides a granular time-series view by recording the available status and *adjusted_price* for each listing on a daily basis. We leveraged this massive dynamic dataset to analyze occupancy trends and to accurately impute missing pricing information in the listings dataset with high precision.

Finally, the Reviews Data (*reviews.csv.gz*) captures historical guest feedback with nearly 1 million entries (986,597 rows). By utilizing fields such as *reviewer_name* and *comments*, this dataset allows us to bridge the gap between static property details and dynamic market behavior, offering a qualitative layer to our quantitative analysis.

Table 1: Dataset Overview & Volume

Dataset	File Name	Description	Volume (Rows)
Listings	*listings.csv.gz*	Static property attributes & location	36,111
Calendar	*calendar.csv.gz*	Daily price & availability history	13,180,517
Reviews	*reviews.csv.gz*	Guest comments & timestamps	986,597

3. Data Preprocessing & Cleaning

Raw real-world data is rarely ready for immediate analysis. We implemented a rigorous cleaning process using the Pandas library [3] to ensure the quality of our dataset and its compatibility with Redis.

Our initial focus was on Data Type Normalization. Since the raw price fields contained currency symbols and commas, numerical aggregation was impossible. We converted these fields into floating-point numbers and standardized missing values (*NaNs*) into safe defaults to prevent insertion errors in our NoSQL database.

The most critical challenge was Handling Missing Price Data. A standard mean imputation would have been inaccurate for a dynamic market like New York. Instead, we engineered a "Smart Imputation Strategy" (detailed in Table 2). This logic prioritizes real historical data from the Calendar dataset over statistical averages, ensuring that our pricing analysis reflects actual market rates rather than broad estimates.

Finally, we performed Feature Selection to optimize memory usage, retaining 10 key attributes: the *listing's id*, *name*, *host_id*, *neighbourhood*, *room type*, *price*, *review count*, *rating score*, and geographic coordinates (*latitude/longitude*) essential for our business queries. A detailed breakdown of the logic applied is provided below.

Table 2: Data Cleaning & Imputation Logic

Step	Operation	Logic Applied
1. Price Cleaning	Type Conversion	Raw price strings are cleaned by removing the '$' symbol and commas, then converted to the Float data type.
2. Primary Imputation	Fill from Calendar Avg	For listings with a missing (*NaN*) price, the average daily price calculated from the granular *calendar.csv* dataset is used as the primary, most accurate source.
3. Secondary Imputation	Fill by Group Median	For any price values still missing after Step 2, they are filled using the median price of the respective *neighbourhood_cleansed* and *room_type* group.
4. Final Fallback	Apply Global Median	In the rare case a price is still null after all group-based logic, it is filled with the global median price of all listings as a last resort.
5. Handle Missing Text	Fill name	Missing values in the listing name field are replaced with the placeholder string "Unknown Listing".
6. Handle Missing Metrics	Fill Reviews & Ratings	Missing values in *number_of_reviews* and *review_scores_rating* are replaced with 0 to allow numerical operations.
7. Feature Selection	Column Reduction	To optimize for memory and query performance, only the 10 key attributes are retained for storage in Redis.

4. Database Design with Redis

Choosing the right data model is critical for performance in any database system. Since Redis is a NoSQL key-value store, mapping relational concepts requires careful planning to ensure efficiency [2]. We selected Redis over traditional SQL databases primarily because of its in-memory architecture which offers sub-millisecond latency. This speed is crucial for real-time applications like a travel booking site where users expect instant search results.

To effectively manage the listing details, we decided to store every property as a Redis Hash. Hashes are the most memory-efficient structure for storing objects with multiple fields such as name, price, and neighborhood. This choice allows us to retrieve or update specific attributes of a listing in constant time

without the need to parse large JSON strings or rewrite the entire object. It ensures that our system remains lightweight even as the number of listings grows.

For efficient filtering and search functionality, we utilized Sets to index listings by their location. We created specific Sets containing listing IDs for each neighborhood, along with a global set for the entire city, to enable both local and aggregate queries. This design choice enables the system to perform fast membership testing and intersection operations. Consequently, queries like finding all listings in Midtown can be executed instantly without scanning the entire database.

We further enhanced the system capabilities by implementing Sorted Sets for ranking purposes. This structure automatically maintains the order of elements based on a score, which in our case was the listing rating. This is the optimal structure for leaderboards as it allows us to query top-rated listings immediately without sorting the entire dataset at runtime. Finally, we utilized Geospatial Indexes to store coordinates with a validation check to ensure that only listings with complete geographical data are indexed. This robust design guarantees the reliability of future location-based queries, such as finding properties within a specific radius.

5. Analysis and Key Insights

Upon populating the Redis database with fresh data, we executed a series of analytical queries to understand the nuances of the New York City rental market. These queries allowed us to move beyond simple statistics and uncover meaningful patterns regarding pricing, location, and guest preferences.

Analysis of Top-Rated Listings

We began by querying the database for listings that achieved a perfect 5.0 rating to identify the characteristics of top-performing properties. The analysis revealed a fascinating dichotomy in the New York market, showing that top-tier ratings are found at both the highest and lowest ends of the price spectrum.

As shown in Figure 1, the upper echelon of the market is dominated by 'Entire home/apt' listings in prime neighborhoods. Properties in Upper East Side, Midtown, and Chinatown command prices between $390 and $629 per night. For instance, the highest-priced 5-star listing (ID: 9879796) in Upper East Side, an entire home/apt titled 'Modern 2 Bedroom by Memorial Sloan Kettering...', is priced at $629, suggesting that a significant segment of travelers equates high quality with exclusivity, privacy, and luxury amenities.

Top 10 Listings Sorted by: Rating (DESC), Price (DESC)

Listing ID	Rating	Price ($)	Room Type	Neighbourhood	Name
9879796	5.00	629.00	Entire home/apt	Upper East Side	Modern 2 Bedroom by Memorial Sloan Kette...
990798417345656604	5.00	614.00	Entire home/apt	Midtown	Beautiful Central Park studio
998088361998367565	5.00	597.00	Entire home/apt	Chinatown	Luxury Penthouse! 2 Bed / 2 Bath + Priva...
986263425862656510	5.00	588.00	Entire home/apt	Hell's Kitchen	Gorgeous space near Hells Kitchen and Mi...
984667555924799835	5.00	526.00	Entire home/apt	West Village	Designer West Village Pool House
994794484302782803	5.00	474.00	Entire home/apt	Hell's Kitchen	Blueground | Midtown W, gym & w/d, nr Hu...
997432199888717593	5.00	419.00	Entire home/apt	Sunset Park	The Greenwood Gem - 3BR
994791843427497764	5.00	407.00	Entire home/apt	Upper West Side	Blueground | UWS, gym, close to central ...
995683781967427706	5.00	391.00	Entire home/apt	Sunset Park	Greenwood Heights Delight - 3BR
986816875695859005	5.00	390.00	Private room	Midtown	The Manhattan Club

Figure 1: Top 10 Highest Rated Listings - Luxury Segment

Conversely, as illustrated in Figure 2, high quality does not always require a high budget. By sorting the 5-star listings by lowest price, we uncovered numerous "Hidden Gems", primarily private rooms in boroughs like Bedford-Stuyvesant and Harlem available for under $50 per night. Listings such as "Bright bedroom with high ceiling in Beds..." (ID: 995268785620684770, $41) prove that budget-conscious travelers can still enjoy top-rated hospitality by choosing residential neighborhoods over tourist hubs.

Top 10 Highest Rated Listings in NYC (Rating DESC, Price ASC)

Listing ID	Rating	Price ($)	Room Type	Neighbourhood	Name (Shortened)
995268785620684770	5.00	41.0	Private room	Bedford-Stuyvesant	Bright bedroom with high ceiling in Beds...
988787058900567282	5.00	41.0	Private room	Bedford-Stuyvesant	Luxury Duplex Apt, HDTV, laundry, terrac...
991301559931083002	5.00	44.0	Private room	Harlem	Beautiful Private room
984866964862347137	5.00	44.0	Private room	Bay Ridge	stylish (balcony)
992033931066536338	5.00	45.0	Private room	Harlem	Affordable private room with shared bath...
995095494292465041	5.00	48.0	Private room	Washington Heights	Comfy Room in Manhattan close to Subway
984670477924074027	5.00	49.0	Private room	Bedford-Stuyvesant	Comfy Bedroom
992285225039563541	5.00	54.0	Private room	Bedford-Stuyvesant	Tuk Ahoy - Sunset Suite (3B)
997093934680613043	5.00	57.0	Private room	Flushing	Lovely Exuberant Home 3A
995686208234684657	5.00	58.0	Private room	Bedford-Stuyvesant	bedstuy is fly(1)

Figure 2: Top 10 Highest Rated Listings - Budget Segment

Neighbourhood Market Dynamics

By aggregating data across different neighborhoods, we identified distinct market segments driven by supply and cost dynamics. Midtown represents the ultra-premium segment, commanding a staggering average price of ~$2,573 per night, making it the most exclusive market in our dataset. Other central hubs like Hell's Kitchen (~$1,391) and Upper West Side (~$1,238) also target high-budget travelers.

In terms of inventory, Bedford-Stuyvesant dominates the market with the highest supply (2,612 listings). It serves as a budget-friendly hub with an average price of just ~$132, offering a sharp contrast to Manhattan's rates. Williamsburg occupies a unique position as a "trendy" middle ground; while it ranks second in supply (2,049 listings), its average price of ~$470 is significantly higher than neighboring Brooklyn areas, reflecting its high demand among younger travelers.

Interestingly, the data reveals an inverse relationship between price and average rating in some hubs.

Harlem ($155 avg price) boasts a notably higher average rating (3.55) compared to the expensive Midtown market (2.58). This suggests that traveler satisfaction is often higher in residential, culturally rich neighborhoods than in purely commercial tourist centers**.**

Table 3: Neighbourhood Statistics - Listings Count, Average Price & Rating

Neighbourhood	Total Listings	Avg Price ($)	Avg Rating
Bedford-Stuyvesant	2,612	$132.18	3.29
Williamsburg	2,049	$470.36	3.46
Midtown	1,993	$2,573.41	2.58
Harlem	1,685	$155.00	3.55
Hell's Kitchen	1,496	$1,391.21	2.98
Upper West Side	1,458	$1,238.80	2.66
Bushwick	1,443	$105.70	3.11
Upper East Side	1,399	$1,051.04	3.11
Crown Heights	1,133	$168.52	3.33
East Village	959	$233.09	3.33

Demand & Availability Trends

We leveraged the granular calendar dataset to analyze daily availability as a direct proxy for booking demand where lower availability indicates higher occupancy. The data reveals a surprising trend that the tightest markets are not the central tourist hubs but rather specific residential and peripheral neighborhoods. As detailed in Table 4 (top 10 most booked neighborhoods), New Dorp stands out as an extreme outlier with 0.00% availability. This indicates that inventory in this area is effectively sold out or completely blocked. Following closely are neighborhoods like Two Bridges and Inwood which both show availability rates near 20%.

Table 4: Top 10 Most Booked Neighbourhoods (Lowest Availability)

Rank	Neighbourhood	Avg Availability (%)	Occupancy Status
1	New Dorp	0.00%	Fully Booked
2	Two Bridges	19.91%	Very High Demand
3	Inwood	20.27%	High Demand
4	Columbia St	20.41%	High Demand
5	Morningside Heights	20.48%	High Demand
6	Little Neck	20.96%	High Demand
7	Riverdale	21.69%	High Demand
8	Civic Center	22.24%	High Demand
9	Battery Park City	22.27%	High Demand
10	Prospect Heights	24.37%	High Demand

The broader trend visualized in Figure 3 suggests that while major hubs like Midtown have a massive supply to absorb tourists, these niche markets face a significant inventory shortage relative to local demand. For hosts, this signals a lucrative opportunity in these specific areas where competition is low and occupancy is nearly guaranteed. The chart below illustrates the top 10 neighborhoods with the lowest availability percentages across New York City.

Figure 3: Top 10 Most Booked Neighbourhoods in NYC

Amenities Preference Analysis

To identify the specific features that drive guest satisfaction, we conducted a text analysis on the amenities field. We filtered exclusively for listings with review scores of 4.5 or higher. This resulted in a robust sample size of 21,367 properties and allowed us to isolate the essential features from the optional ones.

As detailed in Table 5, the data reveals that guest satisfaction is predominantly driven by a mix of safety, connectivity, and practical domestic utility. The two most frequent amenities are "Smoke alarm" and "Wifi" which appear in over 92% of the highest quality properties. This establishes them as non-negotiable baseline expectations where their absence likely disqualifies a listing from achieving a high score.

Table 5: Top 10 Most Common Amenities in High-Rated Listings

Rank	Amenity	Frequency (Count)	Presence (%)
1	Smoke alarm	19,930	93.3%
2	Wifi	19,671	92.1%
3	Kitchen	19,044	89.1%
4	Essentials	18,032	84.4%
5	Carbon monoxide alarm	17,419	81.5%
6	Hangers	16,643	77.9%
7	Hot water	16,141	75.5%
8	Hair dryer	15,349	71.8%
9	Iron	14,931	69.9%
10	Heating	14,765	69.1%

Beyond the top five essentials, the broader analysis visualized in Figure 4 highlights the importance of functional utility items. Features such as "Hangers", "Hair dryer", and "Iron" consistently appear in the top 10 list. This confirms that Airbnb guests prioritize practical domesticity. They prefer a smooth and comfortable residential experience similar to their own home over extravagant hotel luxuries. The chart below illustrates the relative frequency of these top 10 amenities across the dataset.

Figure 4: Top 10 Amenities in Highly Rated Listings

6. Conclusion

This project has successfully engineered a robust and automated Big Data pipeline that transforms raw and voluminous Airbnb data into actionable business intelligence for the New York City rental market. By leveraging the in-memory architecture of Redis, we moved beyond the limitations of static file analysis to create a system capable of delivering real time insights with sub millisecond latency. The implemented pipeline from data ingestion and smart cleaning to efficient Redis data modeling demonstrates a practical and scalable solution for managing dynamic hospitality market data.

The technical implementation confirmed the efficacy of specific Redis data structures for distinct analytical needs. Hashes provided a memory efficient method to store and instantly retrieve detailed attributes for over thirty-six thousand listings. Sorted Sets empowered the system to maintain live rankings of top-rated properties, enabling leaderboard queries without computational overhead. Furthermore, the integration of Geospatial Indexes with validation for data integrity laid a reliable foundation for future location aware searches. These architectural choices collectively ensure the system remains performant and responsive as data volume grows.

From an analytical perspective, the pipeline uncovered nuanced market dynamics that are invaluable for both travelers and hosts. For travelers, the analysis revealed that exceptional guest satisfaction exists across a wide price spectrum, from luxury apartments in Manhattan to budget friendly private rooms in Brooklyn, challenging the assumption that cost alone dictates quality. For hosts, a surprising finding was the extreme occupancy demand in specific residential neighborhoods outside the traditional tourist cores, indicating lucrative and underserved market niches. Moreover, the amenities analysis distilled the essence of a successful listing down to fundamental expectations around safety, connectivity, and practical utility, providing a clear blueprint for hosts to optimize their offerings. For platform administrators, this work validates the use of high performance in memory databases to manage massive datasets with extreme efficiency.

In summary, this work bridges a critical gap between raw data availability and strategic decision making in the sharing economy. It delivers a dual offering comprising a high-performance technical framework for data management and a suite of clear and data driven insights for business strategy. The project validates the power of modern in memory databases like Redis to serve as the engine for real time Business Intelligence, ultimately empowering stakeholders to navigate the complexity of the marketplace with greater speed, efficiency, and confidence.

7. References

[1] Inside Airbnb, "Get the Data - New York City," Inside Airbnb, Oct. 01, 2025. [Online]. Available: http://insideairbnb.com/get-the-data/. [Accessed: Dec. 20, 2025].

[2] Redis Ltd., "Redis Data Types and Commands," Redis Documentation, 2024. [Online]. Available: https://redis.io/docs/.

[3] The Pandas Development Team, "pandas-dev/pandas: Pandas," Zenodo, Feb. 2024. [Online]. Available: https://pandas.pydata.org/.